

ELECTRIC CURRENT OR, ELECTRICITY OR CURRENT ELECTRICITY

- **Introduction of Electricity**

- Electricity is indispensable to us, like as thousands of things we use every day need electricity to work.

For example:-

- ✓ Bulb, T.V, Fan, Motor, A.C, inverter, Cooler, Mobile, Grinder, Iron etc.

- **The father of electricity**

- An English scientist Michael Faraday (1791 – 1867) is known as the father of electricity for his many contributions to the field of electricity.

Note:

- The source of all type of electricity is charge.

- **Charge / Electric Charge / Electrical Charge / Electrostatics Charge**

- Charge is a characteristics of unit of matter that expresses the extent to which it has more or fewer electrons than proton.

Case 1: If no. of protons > no. of electrons, The matter has +ve charge.

Case 2: If the no. of protons < no. of electrons, The matter has –ve charge.

Note:

- If no. of protons = no. of electrons, the matter has no charge. [for e.g atom are electrically neutral because it has equal no. of proton and electron]

- Charge has the basis of all electric phenomena.
- According to Benjamin Franklin, charge has two types.
 - i) Positive charge (+)
 - ii) Negative charge (–)

- Similar charges / Like Charges repel to each – other.

i.e, $\left. \begin{array}{l} (+) (+) \\ (-) (-) \end{array} \right\} \rightarrow \text{Repulsion}$

- Opposite charges or unlike charges attract to each – other.

i.e, $\left. \begin{array}{l} (+) (-) \\ (-) (+) \end{array} \right\} \rightarrow \text{Attraction}$

- The charge of substance depends on the transfer of electrons.
- When two substances are rubbed with each – other, then charge is created on their surfaces.
- In the case of rubbing, the electron is transferred from one substance to another.

Case 1: When a substance loses the electron then it has +ve charge.

Case 2: When a substance gains the electron then it has –ve charge.

For example:-

- ✓ When a glass rod is rubbed with a silk cloth, then glass rod loses the electron and silk cloth gains the electron so,
Silk cloth $\rightarrow$ –ve charge
Glass rod $\rightarrow$ +ve charge

- ✓ When an ebonite rod is rubbed with woollen cloth, the ebonite rod gained electron and woollen cloth lost the electron so,
Woollen Cloth $\rightarrow$ +ve charge
Ebonite rod $\rightarrow$ -ve charge

- The S.I unit of charge is "Coulomb". It is denoted by "C".
- Coulomb named in honour of Charles – Augustin De – Coulomb.
- The unit of charge in C.G.S system is "StatCoulomb".
- 1 StatCoulomb is also called 1 Franklin / 1 e.s.u (electrostatics unit of charge).
i.e, 1 e.s.u = 1 Franklin = 1 StatCoulomb
 $= 3.34 \times 10^{-10}$ Coulomb

- The charge of electron $= -1.6 \times 10^{-19}$ C
The charge of proton $= 1.6 \times 10^{-19}$ C
The charge of neutron = Chargeless

- The smallest unit of charge is "Charge of 1 electron".
i.e, Charge of 1 electron $= 1.6 \times 10^{-19}$ C

↓
The basic unit of charge

- The biggest unit of charge is 'Faraday'.
i.e, 1 Faraday = 96500 C
- The charge is transferable, the process of charge transfer is called 'Conduction'.
- Charge is always associated with mass.
i.e, Charge can't exist without mass but mass can exist without charge.

For example:-

- ✓ Neutrino has no mass can never have a charge.
- In this process of charging the mass of a body doesn't change (a negligible change).

Note:

- Charge is discovered by Benjamin Franklin.

- The charge of a body does not change whatever be its speed accelerated.
- Law of Conservation of Charge
 - "Charge is neither be created nor be destroyed. It can only transfer from one body to another body. i.e, charge is conserved, this law is known as "Law of Conservation of Charge".
- ❖ Charge is quantised, Charge never be fraction whenever a physical quantity have only a distinct value then the quantity is said to be quantised. i.e, Charge is quantised means the charge on a body is multiples of charge of electron (fixed charge).

If the charge on a body = Q

$$\therefore Q = ne$$

where,

$$\begin{aligned} n &= \text{Integer / No. of } e^-(s) \\ e &= \text{Charge of electron} \\ &= 1.6 \times 10^{-19} \text{ C (basic charge)} \end{aligned}$$

Q. How many electrons in 1 Coulomb of Charge?

Ans- Let,

No. of e^- (s) in 1 C = n

Total Charge (Q) = 1 C

Charge of $e^- = 1.6 \times 10^{-19}$ C

$$\therefore Q = ne$$

$$\Rightarrow n = \frac{Q}{e}$$

$$\Rightarrow n = \frac{1 \text{ C}}{1.6 \times 10^{-19} \text{ C}} = \frac{100}{16} \times 10^{18}$$

$$\Rightarrow n = 6.25 \times 10^{18} e^-(s)$$

$$\therefore \boxed{1 \text{ C} = 6.25 \times 10^{18} e^-(s)}$$

Q. Define 1 Coulomb of Charge?

Ans One Coulomb is that quantity of electric charge which exerts a force of 9×10^9 Newtons on an equals charge placed at a distance of 1 metre from it.

$$\begin{aligned} \text{i.e., 1 Coulomb} &= 1 \text{ amp} \times 1 \text{ sec} \\ &= 1 \text{ A} \times \text{S} \end{aligned}$$

The amount of charge that passes through a conductor carrying a current of one ampere in one second.

1 Coulomb of Charge is equal to Charge of $6.25 \times 10^{18} e^-(s)$.

• **Coulomb's Law**

- If the attraction or repulsion force applied between two any objects / bodies is directly proportion to product of charges and inversely proportional to the square of distance between them. The direction of force in the direction of a straight line joining the both charge objects.



Let,

F Force applied between q_1 & q_2

$$\therefore F \propto q_1 q_2 \quad \dots\dots\dots (i)$$

$$\& F \propto \frac{1}{r^2} \quad \dots\dots\dots (ii)$$

On Combining equation (i) & (ii)

$$F \propto \frac{q_1 q_2}{r^2}$$

$$F = k \cdot \frac{q_1 q_2}{r^2}$$

where,

$$K = \frac{1}{4\pi\epsilon_0}$$

K is Coulomb Constant

$$K = 9 \times 10^9 \text{ Nm}^2 / \text{C}^2 \text{ (Approx)}$$

ϵ_0 = Permittivity of Vacuum

$$\epsilon_0 = 8.9 \times 10^{-12} \text{ Fm}^{-1} \text{ (Approx)}$$

Where F is Farad

Note:

- If $q_1 = 1\text{C}$, $q_2 = 1\text{C}$ & $r = 1\text{m}$.

then,

$$F = k \cdot \frac{q_1 q_2}{r^2}$$

$$\Rightarrow F = k \times \frac{1 \times 1}{1^2}$$

$$\Rightarrow F = k = 9 \times 10^9 \text{ N (Approx)}$$

- Electric Current**

➤ The rate of flow of electric charge particles through any conductor is called electric current.

OR

The rate of flow of charge through high potential to low potential in any conductor is called electric current.

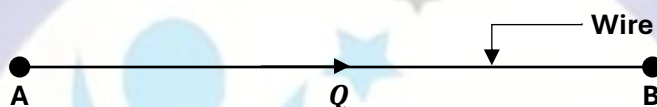
Note:

➤ The rate of flow of charge means the rate of flow of electrons.

➤ It is a scalar quantity.

➤ It is denoted by 'I'.

➤ If 'Q' charge flows through an electric wire up to time 't' then the electric current passing through this wire is



$$\therefore \text{Electric Current} = \frac{\text{Amount of charge}}{\text{Taken Time}}$$

$$\therefore I = \frac{Q}{t}$$

$$\Rightarrow Q = I \cdot t$$

$$\Rightarrow t = \frac{Q}{I}$$

where,

Q = amount of charge

t = time taken

$$\rightarrow \text{The S.I unit of electric current} = \frac{\text{the S.I unit of charge}}{\text{the S.I unit of time}} = \frac{C}{S}$$

Note:

➤ 1 C / S = 1 ampere.

i.e, the S.I unit of electric current is "ampere". It is denoted by "A".

→ "Ampere" named in honoured of French Scientist "Andre Marie Ampere".

- Defined One ampere electric current**

➤ If one coulomb of charge passes through a wire in one second, then the amount of current is 1 ampere.

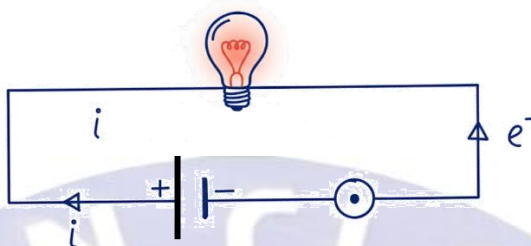
OR

If $6.25 \times 10^{18} \text{ e}^-$ (s) passes through one end of electric wire to another end in one second, called "1 ampere".



Note:

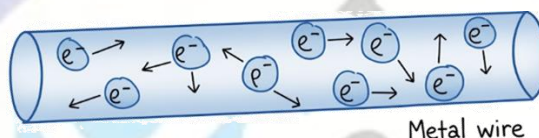
- The conventional direction of electric current from positive ends to negative ends.



But the electrons move negative ends to positive ends (terminal).

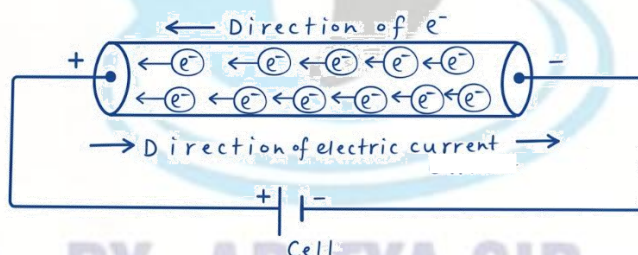
▪ **How the electric current flows in a wire**

Case 1: When no current passes through a conductor



The electrons present in it move at random in all the direction between the atoms of the metal wire.

Case 2: When the cell is connected between ends of metal wire. i.e, passes electricity through the wire.



In this case, an electric force acts on the electrons present in the wire. Since electrons are negatively charged, they start moving from negative end to the positive ends of the wire.

• **Source of electricity**

- Solar energy, water energy, wind energy, chemical energy (cell), mechanical energy (generator), Heat energy, nuclear energy etc.

▪ **Electricity can be classified into two parts:**

1. Static electricity.
2. Current electricity.

1. **Static electricity**

- That electricity in which charge can not move i.e, charge is in rest, called Static electricity.

For example:-

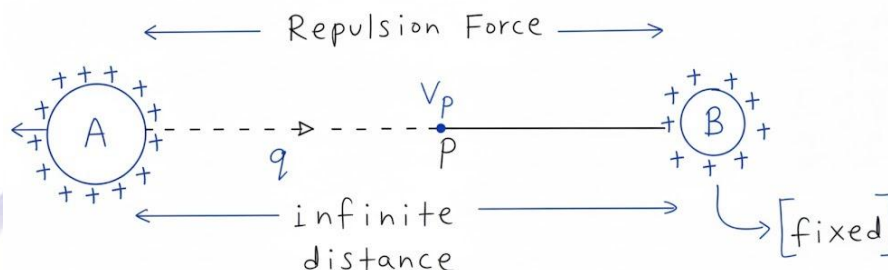
- ✓ The charge on glass rod when it is rubbed with silk cloth.
- ✓ The charge on the comb when it is rubbed with hair.

2. Current electricity

- That electricity in which charge can move from one end to another end is called current electricity.

For example:-

- ✓ The electricity which uses in houses, schools, building, office etc.
- **Electric potential / Potential**
 - The electric potential at a point is defined as, The magnitude of work done in moving a unit charge from infinity to that point.



→ If the work done in moving q charge from point B to P is W.

∴ By the definition of electric potential.

$$\text{Electric potential} = \frac{\text{Work done in moving charge}}{\text{Amount of charge}}$$

$$\therefore V = \frac{W}{q}$$

→ The S.I unit of electric potential = $\frac{\text{The S.I unit of Work}}{\text{The S.I unit of Charge}} = \frac{J}{C}$

Note:

➤ 1 J / C = 1 volt = 1 V

→ Symbol

- The S.I unit of electric potential is 'Volt'.

Named in honor of the Italian Scientist
"Alessandro Volta".

- The electric potential of earth is '0'.
- The electric potential of uncharged substance is zero (0).
- The electric potential of negatively charged substances is negative.
- The electric potential of positively charged substances is positive.

- **Define One volt**

- When one joule of work done is moving in unit positive charge from infinity to that point is known as one volt.

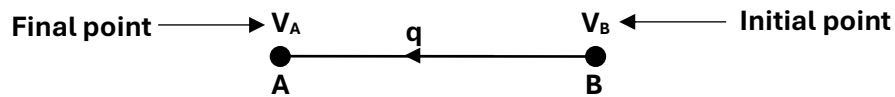
i.e, 1 volt = 1 j / C

Note:

- Electric potential is denoted by 'V' because electric potential is also called 'voltage'.

- **Potential Difference**

- The potential Difference between two points is defined as, "The magnitude of work done in moving unit charge from one point to another point."



$$\text{Potential Difference} = \frac{\text{Work done}}{\text{Amount of moving (Charge)}}$$

$$\Rightarrow V_{AB} = V_A - V_B = \frac{W_{AB}}{q}$$

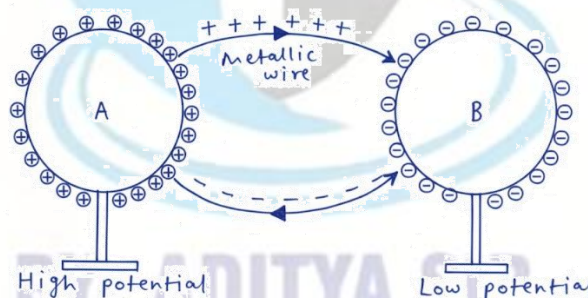
- It is a scalar quantity.
- The S.I unit of potential difference is 'Volt' which is denoted by 'V'.

$$V_{AB} = \frac{W_{AB}}{q}$$

Potential Difference between A & B

$$\Rightarrow V = \frac{W}{q} \quad \Rightarrow W = V \cdot q$$

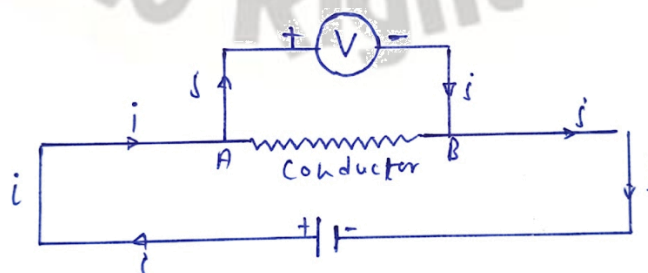
- The free negative charge moves from lower potential to higher potential.
- The free positive charge moves from high – potential to low potential.
- There is an experiment.



- Potential difference is also called 'Voltage'.

- **Voltmeter**

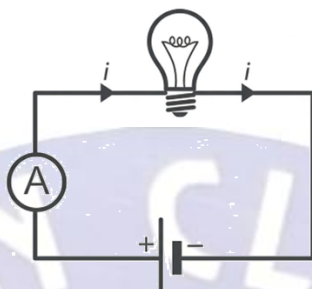
- Voltmeter is that electric device measures potential difference between two points in a circuit.
- Voltmeter is connected in parallel connection with the conductor.
- A voltmeter has a high resistance due to high resistance it takes a negligible current from the circuit.



- The resistance of an ideal voltmeter is infinite.
- Its symbol is

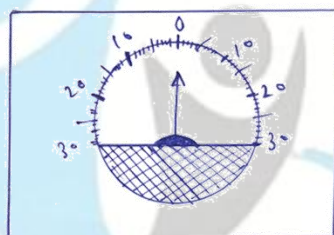
- **Ammeter**

- Ammeter is that electric device which measures electric current in a circuit.
- Ammeter is connected in series connection with the circuit.
- It has a very low resistance due to this reason there are no change in the current.
- Its symbol is 



- **Galvanometer**

- A galvanometer is an instrument that can detect the presence of a current in a circuit it is connected in parallel connection.
- Its symbol is 



Galvanometer

- It has a pointer. If it remains at zero than zero current flowing through it.
- It can deflect either to the left or to the right of the zero mark depending on the direction of electric current.

- **Cell / Dry Cell / Voltian Cell**

- Cells are those electric device which provide constant potential difference between the both electrode of cells by the chemical reaction in the cell.
- That cell which produces constant electric current, called voltian cell or dry cells.
- In the dry cell, the higher potential is known as positive terminal and lower potential is known as negative terminal.



For example:-

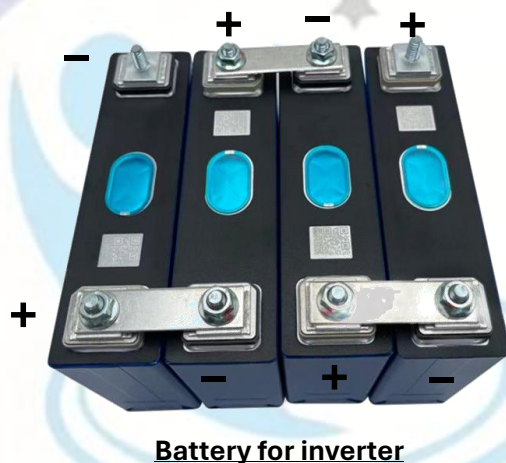
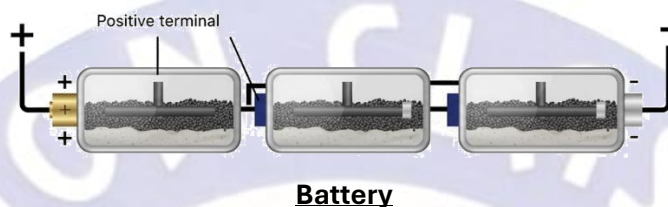
- ✓ The cells which are used in torch, radio, wall watch, walk man etc.

Note:

- Dry cells convert chemical energy in electrical energy by presence of electrode in wire.

• **Battery**

- The combination of cells (two or more) is called battery.



• **Electric Circuit**

- That closed path in which a current can flow is called an electric circuit.

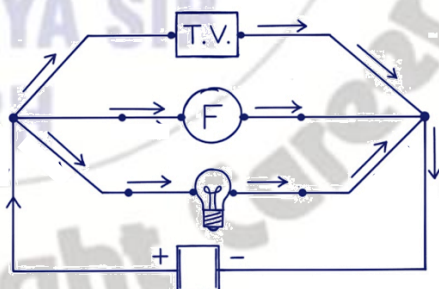
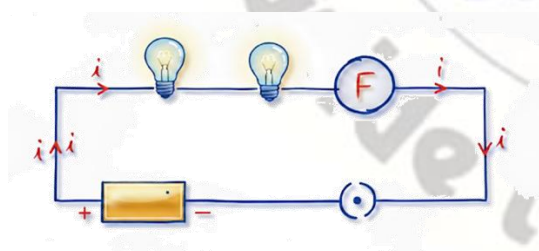


Diagram of Electric Circuit

▪ **Some elements of electric circuit and their symbols**

	Electric Current	Symbol
1.	An electric cell	
2.	Combination of cells OR battery	
3.	Plug Key (Open)	
4.	Plug Key (Close)	

5.	A wire joint	
6.	Wire Crossing without joining	
7.	Electric bulb	
8.	A fixed Resistance OR resistor	
9.	Open Switch	
10.	Closed Switch	
11.	Ammeter	
12.	Voltmeter	
13.	Galvanometer	
14.	Electric fuse	
15.	Variable resistance OR Rheostat	

• Ohm's Law

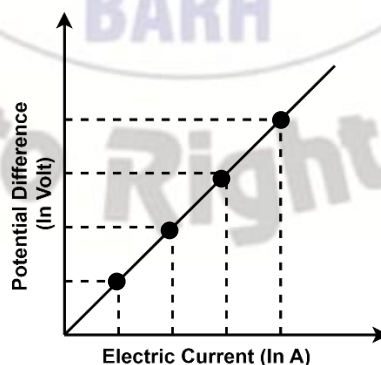
- In 1826, first established by the German Scientist "George Simon Ohm".
- According to George Simon Ohm, The current through the metallic element is proportional to the potential difference applied between its ends, provided the temperature remains constant.
- If the potential difference 'V' is applied to an element and a current i passes through it.

It can be written in the form of

$$V \propto I \text{ (When temp is constant)}$$

$$\Rightarrow V = I \cdot R,$$

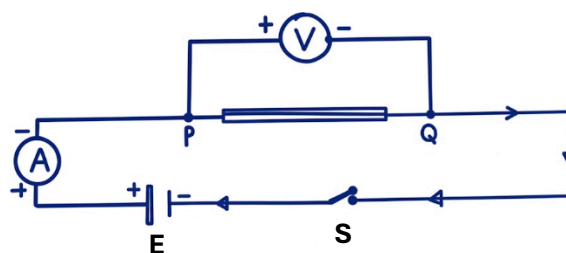
Where, R is a constant of that element is called resistance.



The electric current and potential difference diagram.

▪ Experiment to verify Ohm's Law

- First we take a dry cell (E), an ammeter (A), a voltmeter (V), a switch (S), and a nichrome wire PQ are connected in given diagram.



- When switch put on then electric current is flowing in the circuit ammeter (A) measures the current I , Voltmeter (V) measures the potential difference between the points P and Q of nichrome wire PQ. Now, put two cells in the place of one cell and we note the reading of V / I , again, we used four cells, five cells, ... and after in all cases we have a constant value of V / I . Which shows that electric current is directly proportional to potential difference between ends.

Note:

- This law is not applicable for transistor, Thermistor, diode etc. because it is heat during flow electric current.

- **Resistance**

- That property of any conductor which opposes the flow of current through it is called resistance.

By Ohm's Law

$$V = I \cdot R$$

$$\text{OR, } R = \frac{V}{I}$$

$$\Rightarrow \text{Resistance} = \frac{\text{Potential Difference}}{\text{Electric current}}$$

- The S.I unit of resistance is Ohm. It is denoted by ' Ω '.

$$\therefore 1 \text{ Ohm } (\Omega) = \frac{1 \text{ Volt}}{1 \text{ Ampere}}$$

$$\text{OR, } 1 \Omega = 1 \text{ V / A}$$

- **Define 1 ohm (Ω) Resistance**

- If a potential difference of 1 volt is applied across an element and a current of 1 ampere passes through it, the resistance of the element is called 1 ohm (1Ω).

- **Why do resistances create in the conductor?**

- The electric current is a flow of electrons through a conductor, when the electrons move from one part of the conductor to the other part. They collide with other electrons and with the atoms and ions present in the body of the conductor due to these collisions, there is some opposition to the flow of electron through the conductor. This opposition or disturbance is called resistance of that conductor.

Note:

- Insulators have infinite resistance.

- The Resistance of a conductor depends on length, thickness, nature of material and temperature of the conductor.

1. **Depends on the length of wire**

- The resistance of the wire is directly proportional to its length.

$$\text{Resistance} \leftarrow R \propto l \rightarrow \text{length of wire} \quad \text{..... (i)}$$

2. Depends on the cross – section area (thickness) of wire

- The resistance of the wire is inversely proportional to its thickness.

$$R \propto \frac{1}{A} \quad \text{..... (ii)}$$

└───────────> Cross – section of area

3. Depends on the nature of material

- The resistance of different materials having same length and thickness are different.

For example:-

- ✓ The resistance of silver and copper of same length and thickness are different.

4. Depends on the temperature

- The resistance of conductor is directly proportional to the temperature.

$$R \propto T$$

└───────────> Temperature

Now on combining equation (i) and (ii)

$$R \propto \frac{l}{A}$$

$$\Rightarrow \boxed{R = \rho \cdot \frac{l}{A}}$$

Where, ρ (Rho) is a constant which is called resistivity of wire.

OR,

$$\text{The resistivity } (\rho) = \frac{R \cdot A}{l}$$

$$\begin{aligned} \rightarrow \text{The S.I unit of resistivity} &= \frac{\text{The S.I unit of } R \times \text{The S.I unit of } A}{\text{The S.I unit of length}} \\ &= \frac{\Omega \cdot \text{m}^2}{\text{m}} = \Omega \cdot \text{m} \end{aligned}$$

i.e, The S.I unit of resistivity (ρ) is ' $\Omega \cdot \text{m}$ '.

Note:

1. If length of wire (l) = 1m
 area of wire (A) = 1m²

$$\begin{aligned} \therefore \rho &= \frac{R \cdot A}{l} \Rightarrow \rho = \frac{R \times 1^2}{1} \\ &\Rightarrow \rho = R \end{aligned}$$

i.e, the resistivity is numerically equal to the resistance of wire.

2. The resistivity of a substance is a material property. It does not depends on length and thickness of wire. It depends on the nature of substance and temperature.

▪ **Some substances and their resistivity**

Materials	Resistivity ($\Omega \text{ m}$)
Type – (1) metals	
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminum	2.8×10^{-8}
Tungsten	5.8×10^{-8}
Iron	9.7×10^{-8}
Type – (2) Alloy	
Manganin { Cu = 84% Mn = 12% Ni = 4%	48.2×10^{-8}
Nichrome { Ni = 60%, Cr = 12% Mn = 2%, Fe = 26%	100×10^{-8}
Type – (3) Semi – Conductor	
Germanium	4.7×10^{-3}
Silicon	2.5×10^3
Type – (4) Insulator	
Diamond	5×10^{12}
Fused quartz	$10^{16} - 10^{19}$

• **Conductors**

- Those conductors which have very low electrical resistance are called conductors.

OR

Those substances which allow to flow electric current through them, called conductors.

For example:-

- ✓ Silver, Copper, Aluminum, Iron etc.

Note:

- Conductors having free electrons due to which, conduct electricity.

• **Good Conductors**

- Those conductors which have very low electrical resistance are called good conductors.

For example:-

- ✓ Silver, Copper, etc.

• **Insulator**

- Those substances which infinitely high resistance are called insulators.

For example:-

- ✓ Wood, Bricks.

Note:

- Insulators have no free e^{-} (s).

• **Resistors**

- Those conductors which have high resistance are called resistors.
 ➤ Resistors have high melting points.

For example:-

- ✓ Nichrome wire, electric iron, electric heater, water rod, bulb.

- **Semi – Conductors**

- The resistivity of the semi – conductors between the good conductors and insulator are known as semi – conductors.
- Semi – conductors are used to make integrated circuits (ics) in computers, T.V, modern telephones etc.

For example:-

- ✓ Silicon, germanium, boron etc.

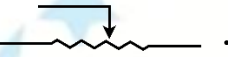
- **Super – Conductors**

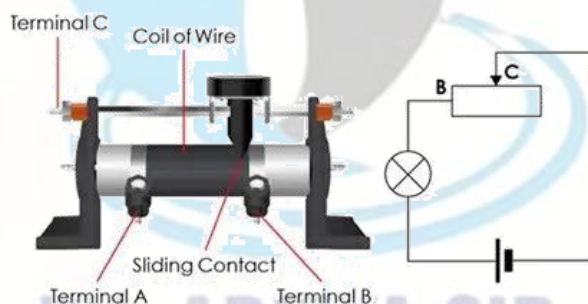
- Those conductors which have a negligible resistance are called super – conductors.

For example:-

- ✓ When good conductors kept below to 4K.

- **Rheostat**

- The variable resistances are also known as Rheostat.
- The symbol of Rheostat  OR .
- It is used to change the electric current in the circuit without changing the voltage source like as cells or, battery.
- It is used for Motor speed control, Lighting control, heater, Oven, Temperature control etc.



Rheostat

- **Conductivity**

- The conductivity of materials is defined as, “It is the material’s ability to conduct electricity.”
- Electric current can flow easily through a material with high conductivity.
- The S.I unit of conductivity is Siemen / meter (S / m).
- Conductivity is denoted by σ (mho).

↑
Small Sigma

- **Combination / Connection of resistors**

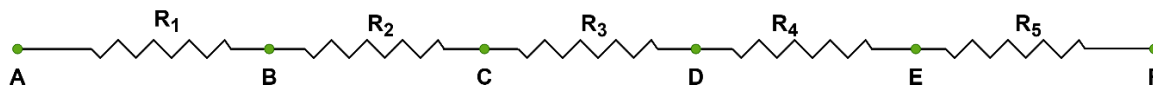
- When two or more resistors are connected by a particular way then, this way is called combination of resistors.

- **There are three types of combination of resistors**

1. Series combination / Connection.
2. Parallel combination / Connection.
3. Mixed Combination / Connection.

1. Series combination

- That combination of resistors in which resistors are connected end to end consecutively, called series combination.



For example:-

- ✓ The series combination of bulbs are used in decoration of light in our house building, Deepawali, Durga puja etc.



• Equivalent Resistance / Resultant Resistance

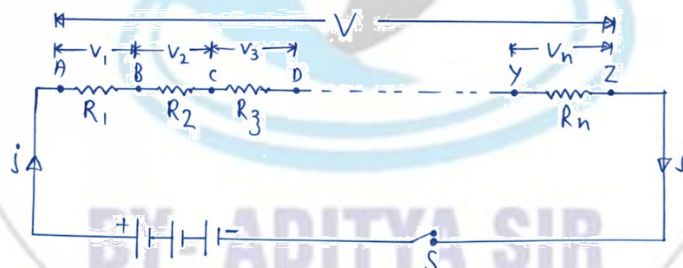
- An equivalent resistance, that is equivalent to the combination of all resistances.
 OR
 ➤ A single resistance which is equivalent to the combination of all resistances, called equivalent resistances.
 ➤ An equivalent resistance it denotes by 'Req'.

Q. Find the equivalent resistance in series combination.

OR

An expression for equivalent resistance in series combination.

Ans Let n resistors of resistances $R_1, R_2, R_3, \dots, R_n$ are connected in series combination.



If the potential difference between points AB, BC, CD, ..., YZ are $V_1, V_2, V_3, \dots, V_n$ respectively.

Let total electrical potential difference between points A & Z is V.

And equivalent resistance in the circuit is R_{eq} / R_s .

$$\therefore V = V_1 + V_2 + V_3 + \dots + V_n \quad \dots\dots\dots (i)$$

By ohm's law,

$$V = I R_{eq}$$

$$V_1 = I R_1$$

$$V_2 = I R_2$$

$$V_3 = I R_3$$

$$\dots\dots\dots$$

$$\dots\dots\dots$$

$$V_n = I R_n$$

Putting the value of $V_1, V_2, V_3, \dots, V_n$ in equation (i)

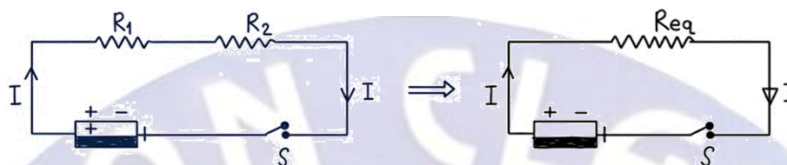
$$\Rightarrow I \cdot R_{eq} = I R_1 + I R_2 + I R_3 + \dots + I R_n$$

$$\Rightarrow \cancel{I} \cdot R_{eq} = \cancel{I} (R_1 + R_2 + R_3 + \dots + R_n)$$

$$\Rightarrow \boxed{R_{eq} = R_1 + R_2 + R_3 + \dots + R_n}$$

→ Which is required expression

⇒ Let 2 resistors of resistances R_1 & R_2 are connected in series connection.



$$\therefore R_{eq} = R_1 + R_2$$

Note:

1. If $R_1 = R_2 = R_3 = \dots = R_n = r$

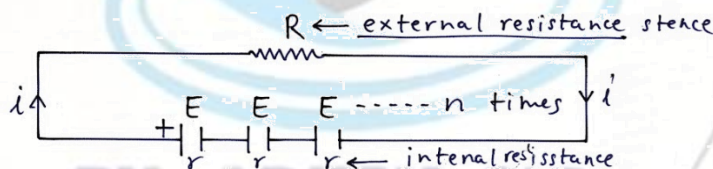
$$\text{Then, } R_{eq} = R_1 + R_2 + R_3 + \dots + R_n \\ = r + r + r + \dots + r$$

$$\Rightarrow R_{eq} = r \times n$$

2. In series connection the electric current is constant for every resistance.

3. In series connection the value of electric potential is different for different resistance.

⇒ Find electric current in series combination / connection.



$$\therefore \text{Electric current} = \frac{\text{Total e.m.f (V)}}{\text{Total resistance}}$$

$$\text{OR, } I = \frac{V}{R}$$

$$\therefore \boxed{I = \frac{nE}{R + nr}}$$

where,

n = no. of e.m.f

= no. of internal resistance

R = external resistance

Case 1: If $R \gg nr$

→ much greater than

then, $R + nr \approx R$

→ Approx

$$\therefore I = \frac{nE}{R + nr}$$

$$\therefore I = \frac{nE}{R}$$

→ which is required I

Case 2: If $R \ll nr$

→ much less than

$$\therefore R + nr = nr$$

$$\therefore I = \frac{nE}{R + nr}$$

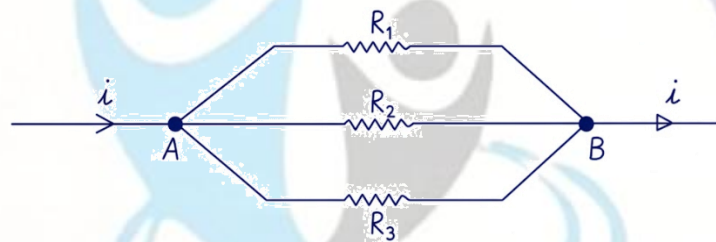
$$\therefore I = \frac{nE}{nr}$$

$$\Rightarrow I = \frac{E}{r}$$

→ which is required I

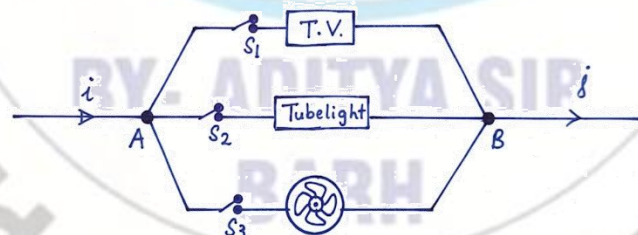
2. Parallel Combination / Connection

- That combination in which two or more resistors are connected between the same two points, parallel combination.



For example:-

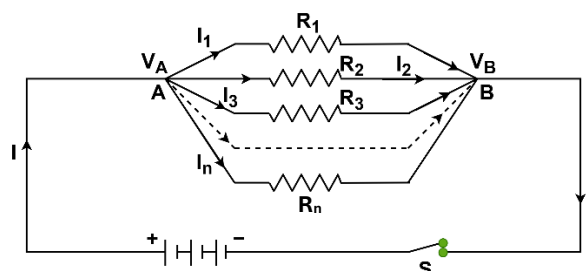
- ✓ The parallel combination of resistors are used in our homes, office, school, etc.



• Equivalent resistance in parallel combination

- Let n resistors of resistances $R_1, R_2, R_3, R_4, \dots, R_n$ are connected in parallel connection between two points A & B.

Let equivalent resistance of circuit is R_{eq} / R_p .



Let potentials on points A and B are V_A and V_B .

Hence, the potential difference between points A & B is $V_A - V_B$.

If battery supplied, I current which are distributed / isolated on point A in I_1, I_2 & I_3 and again they combined with each other at point B and formed current.

Thus, $I = I_1 + I_2 + I_3 + \dots + I_n$ (i)

By Ohm's law,

$$I = \frac{V_A - V_B}{R_{eq}}$$

Similarly,

$$I_1 = \frac{V_A - V_B}{R_1}$$

$$I_2 = \frac{V_A - V_B}{R_2}$$

$$I_3 = \frac{V_A - V_B}{R_3}$$

$$I_n = \frac{V_A - V_B}{R_n}$$

Putting the value of $I, I_1, I_2, I_3, \dots, I_n$ in eq (i)

$$\Rightarrow \frac{V_A - V_B}{R_{eq}} = \frac{V_A - V_B}{R_1} + \frac{V_A - V_B}{R_2} + \frac{V_A - V_B}{R_3} + \dots + \frac{V_A - V_B}{R_n}$$

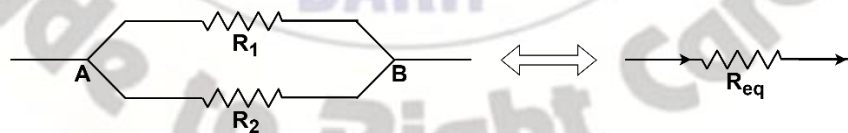
$$\Rightarrow \frac{V_A - V_B}{R_{eq}} = V_A - V_B \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n} \right)$$

$$\Rightarrow \boxed{\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}}$$

→ Which is required equivalent resistance

Case 1:

➤ Let two resistors of resistances R_1 and R_2 are connected in parallel connection then R_{eq} .



$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 \cdot R_2}$$

$$\Rightarrow R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

Case 2:

➤ When three resistors of resistances R_1, R_2 and R_3 are connected in parallel combination then

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{R_2 \cdot R_3 + R_1 \cdot R_3 + R_1 \cdot R_2}{R_1 \cdot R_2 \cdot R_3}$$

$$\therefore \boxed{R_{eq} = \frac{R_1 \times R_2 \times R_3}{R_1 \cdot R_2 + R_2 \cdot R_3 + R_3 \cdot R_1}}$$

Note:

1. In parallel connection, there is no change in voltage but electric current is different for different resistance.

2. If $R_1 = R_2 = R_3 = \dots = R_n = r$

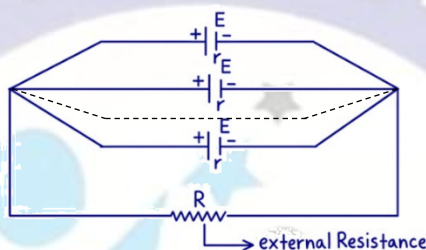
$$\therefore \frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \dots + \frac{1}{r}$$

$$\Rightarrow \frac{1}{R_{eq}} = n \times \frac{1}{r}$$

$$\Rightarrow R_{eq} = \frac{r}{n}$$

$\Rightarrow$ Find Electric current in the circuit.



$$\therefore I = \frac{V}{R} = \frac{\text{Total e.m.f}}{\text{Total resistance}}$$

$$\Rightarrow I = \frac{E}{R + \frac{r}{n}} = \frac{E}{\frac{Rn + r}{n}} = \frac{nE}{Rn + r}$$

$$\Rightarrow I = \frac{nE}{Rn + r}$$

Where, n = no. of shells
 = no. of internal resistance

Note:

1. If $R \gg \frac{r}{n}$
 and $R + \frac{r}{n} = R$

$$\therefore I = \frac{E}{R}$$

2. If $R \ll \frac{r}{n}$

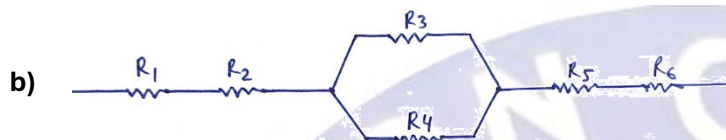
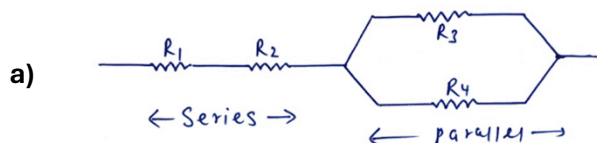
then, $R + \frac{r}{n} = \frac{r}{n}$

$$\therefore I = \frac{E}{\frac{r}{n}} = \frac{nE}{r}$$

$$\Rightarrow I = \frac{nE}{r}$$

3. Mixed Combination / Connection

- When two or more resistors are connected series as well as parallel combination is called mixed connection.



- **Equivalent resistance in mixed combination**

- The equivalent resistance in mixed combination is defined as, "It is a single resistance which is equal to arithmetic sum of equivalent resistance of series and parallel connections".

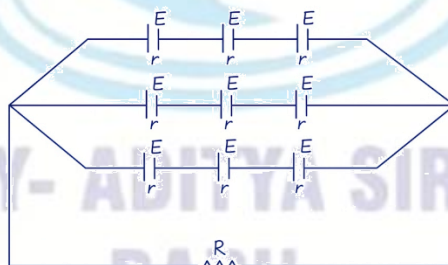
For example:-

- ✓ The equivalent resistance in Q.N

i) $R_{eq} = R_1 + R_2 + \frac{R_3 \times R_4}{R_3 + R_4}$

ii) $R_{eq} = R_1 + R_2 + \frac{R_3 \times R_4}{R_3 + R_4} + R_5 + R_6$

- **Finding electric current in mixed combination in special case**



$$\because V = IR \quad \Rightarrow I = \frac{V}{R}$$

$$\because I = \frac{\text{Total e.m.f}}{\text{Total resistance}}$$

$$\Rightarrow I = \frac{nE}{R + \frac{nr}{m}} = \frac{nE}{\frac{mR + nr}{m}} = \frac{mnE}{mR + nr}$$

$$\therefore \boxed{I = \frac{mnE}{mR + nr}}$$

where,

m = no. of rows

n = no. of shells

n = no. of internal resistances

Note:

- The value of I is maximum when the value (mR + nr) is minimum.

$$\begin{aligned}\therefore mR + nr &= (\sqrt{mR})^2 + (\sqrt{nr})^2 \\ &= (\sqrt{mR} - \sqrt{nr})^2 + 2\sqrt{mR \times nr}\end{aligned}$$

Hence, the value of $mR + nr$ will be minimum only when the value of $(\sqrt{mR} - \sqrt{nr})^2$ will be zero.

$$\Rightarrow (\sqrt{mR} - \sqrt{nr})^2 = 0$$

$$\Rightarrow \sqrt{mR} - \sqrt{nr} = 0$$

$$\Rightarrow \sqrt{mR} = \sqrt{nr}$$

$$\Rightarrow \boxed{R = \frac{nr}{m}}$$

→ The value of R gives maximum value of I.

• Heat Produced by Current

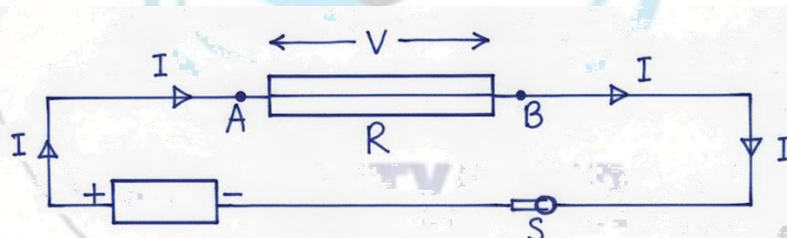
- When the electric current flows through a resistor, then electrical energy gets converted into heat energy which is known as heat produced by electric current.
- The heat produced by electric current is denoted by 'H'.

Note:

1. If the same current i passes through two resistances in a given time, more heat will be produced in the larger resistances.
2. The heat is produced by electric current is also called 'joule heating'.

▪ The amount of heat produced by electric current.

- Let I current passes through a conductor of resistance R , for a time t .
- If the potential difference between ends of conductor is 'V'.



$$\therefore V_1 = \frac{W}{Q} \quad \Rightarrow W = V \cdot Q$$

- The amount of work done in moving Q charge from A to B is W . This work done is equal to Heat produced by electric current.

$$\therefore H = Q \cdot V$$

$$\Rightarrow H = I \cdot t \times I \cdot R \quad (\because Q = I \cdot t \text{ \& } V = I \cdot R)$$

$$\Rightarrow H = I^2 \cdot R \cdot t$$

$$\Rightarrow H = I^2 \cdot R \cdot t$$

→ This is known as "Joule's equation of electrical heating".

Note:

- Heat Produced (H) by electrical current depends on three factors.

$$\left. \begin{array}{l} \text{a) } H \propto I \\ \text{b) } H \propto R \\ \text{c) } H \propto t \end{array} \right\} \therefore H = I^2 \cdot R \cdot t$$

→ longer the time of flow of electric current

$$\begin{aligned}\therefore H &= I^2 \cdot R \cdot t \\ &= \left(\frac{V}{R}\right)^2 \times R \times t \quad (\because I = \frac{V}{R}) \\ &= \frac{V^2}{R^2} \times R \times t = \left(\frac{V^2}{R}\right) \times t\end{aligned}$$

$$\therefore \boxed{H = \left(\frac{V^2}{R}\right) \times t}$$

OR, $H = Q \cdot V = I \cdot t \cdot V$ ($Q = I \cdot t$)

$$\Rightarrow \boxed{H = V \cdot I \cdot t} \quad \Rightarrow \boxed{H = \frac{V^2 t}{R}}$$

The S.I unit of Produced Heat is 'Joule'.

Note:

$$\begin{aligned}\therefore 1 \text{ Calorie} &= 4.182 \text{ Joule} \\ &\approx 4.2 \text{ Joule}\end{aligned}$$

$$\therefore 1 \text{ Joule} = \frac{1}{4.2} \text{ Calorie}$$

$$\therefore W \text{ Joule} = \frac{W}{4.2} \text{ Calorie}$$

$\therefore$ Heat Produced in t time (H) =

$$\begin{aligned}H &= \frac{W}{4.2} \text{ Calorie} \\ \Rightarrow H &= \frac{V \cdot I \cdot t}{4.2} \text{ Calorie} \\ \Rightarrow H &= \frac{V^2 \cdot t}{4.2 \times R} \text{ Calorie} \\ \Rightarrow H &= \frac{I^2 \cdot R \cdot t}{4.2} \text{ Calorie}\end{aligned}$$

Special Note:

➤ If voltage and time is constant.

then, $H \propto \frac{1}{R}$ ($\because H = \left(\frac{V^2}{R}\right) \cdot t$)

i.e, If the same potential difference is applied across two resistances, more heat will be produced in the smaller resistance.

For example:-

- ✓ If two resistors of resistances. 5Ω & 10Ω is applied across 220 V then more heat will be produced by 5Ω resistance.

• **Electric / Electrical Power**

➤ The rate of electrical work done is called electric power.

OR

➤ The rate of which electrical energy is consumed is called electric power.

$$\therefore \text{Electrical power} = \frac{\text{Electrical Work done}}{\text{Time taken}}$$

➤ Electric Power is denoted by 'P'.

$$\therefore P = \frac{W}{t} = \frac{U}{t} = \frac{H}{t}$$

- The S.I unit of electrical power = $\frac{\text{The S.I unit of } H}{\text{The S.I unit of Time}} = \text{Joule / Second} = \text{J / S}.$

Note:

- 1 J/S = 1 watt
 Named in honored of 'James Watt'.

- The smallest unit of electrical power is watt and the biggest unit kilowatt (Kw).
 1 Kilowatt (1 Kw) = 1000 W

• **Define 1 Watt**

- When an electric appliance consumes electric energy at the rate of one Joule per second then we called the power of appliance is 1 watt.
 i.e, 1 W = 1 J / S

$$\therefore P = \frac{H}{t} = \frac{I^2 \cdot R \cdot t}{t}$$

$$\Rightarrow P = I^2 \cdot R$$

$$\Rightarrow P = I \cdot I \cdot R$$

$$\Rightarrow P = I \cdot V \quad (\because V = I \cdot R)$$

$$\Rightarrow P = V \cdot I$$

$$\Rightarrow P = V \times \frac{V}{R} \quad (\because I = \frac{V}{R})$$

$$\Rightarrow P = \frac{V^2}{R}$$

▪ **The electric power depends on three factors**

i) $P \propto I$

$$\because P = V \cdot I$$

ii) $P \propto R$

$$\because P = I^2 \cdot R$$

→ When I is constant.

iii) $P \propto \frac{1}{R}$

$$(\because P = \frac{V^2}{R})$$

→ When V is constant.

iv) $P \propto V$

$$(\because P = V \cdot I)$$

Note:

- Electric power is directly proportional to V & I but neither directly nor inversely proportional to R. it is unconditional.

- Other unit of electric power is Horse Power (HP).

Note:

- 1 Horse Power (H.P) = 746 w.

Special Note:

1. Higher the power of electrical appliances, lower their resistance (when v is constant).

For example:-

- ✓ Two bulbs of rating (100 w, 220 V) & (60 w, 220 V), The 100 w bulb has low resistance.
2. The (100 w, 220 V) has more much bright than the bulb (60 w, 220 V).

In both cases the voltage is constant.

- **Electric / Electrical Energy**

- Electric energy is defined as, "The product of power rating and time used by it of electric appliance".
- It is denoted by 'E'.

$$\therefore \text{Electric energy (E)} = \text{Power} \times \text{time}$$
$$\Rightarrow E = P \times t$$

- The S.I unit of electrical energy is 'Joule'.

- **Electric energy depends on only two factors**

- i) $E \propto P$
- ii) $E \propto t$

Note:

- The biggest unit of electric energy is kilowatt hour (KwH).
 $\therefore 1 \text{ KwH} = 1000 \text{ W Hour}$

- **Commercial Unit of Electrical energy**

- 1 KwH / 1 Commercial Unit / 1 Unit / 1 B.O.T Unit is called commercial unit of electrical energy.

1 KwH is defined as, "The amount of electrical energy consumed when an electrical appliance having a power rating of 1 Kw is used for 1 Hour.

$$\begin{aligned}\therefore 1 \text{ KwH} / 1 \text{ Unit} &= 1000 \text{ w} \times \text{Hour} \\ &= 1000 \text{ J/S} \times 3600 \text{ S} \\ &= 1000 \times 3600 \text{ J} \\ &= 3600000 \text{ J}\end{aligned}$$

$$\therefore 1 \text{ KwH} = 1 \text{ Unit} = 3.6 \times 10^6 \text{ J}$$

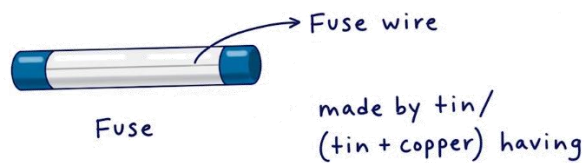
- **Uses of Heating effect of Electric current**

- Electric bulbs, Rooms heater, blowers, electric irons, immersion heater, toaster, electric fuse etc.

- **Electric Fuse**

- An electric fuse is a safety electric device that allow excessive amount of electric current flow through an electric circuit is consists of a metallic wire of low melting point, fixed between the two terminals of a fuse plug. The fuse plug fits into a fuse socket connected in the circuit. In the case of high electric current or overloading the fuse wire gets heated too much, melts and break the circuit due to which the current stops flowing. This prevents the fire in the house and damage to various electrical appliances.

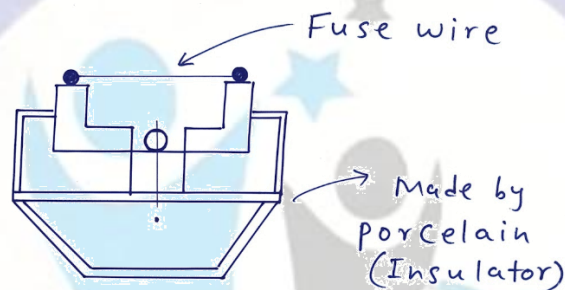
Type 1:



Capacity → 5 A to 15 A

Used in → T.V, inverter, Stabilizer etc.

Type 2:



Capacity → 15A to ∞

Used in → Household wire & others

Note:

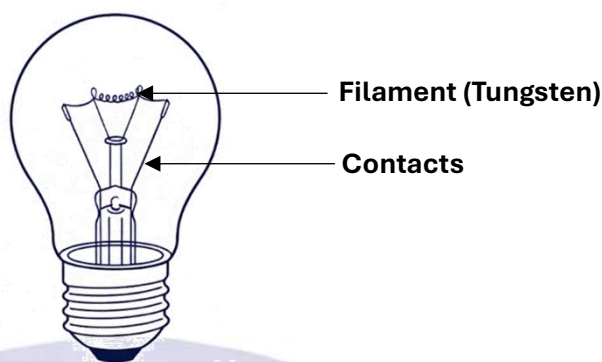
1. All type of fuses are connected in series connection with red wire (main wire) in the circuit.
2. The thickness of fuse wire depends on load of electric appliances used.
3. The heavy fuse which is used buildings, factory, offices, having capacity 32A, 63A, 120A.

▪ **Working rule of fuse**

- We know that electric fuse is connected in series with an appliance, so the current through the fuse is same of the current through the appliance / group of appliances. If the current exceeds the safe value, the heat produced in the fuse wire caused it to melt immediately. This breaks the circuits, preventing any damages of appliances.

• **Electric bulb**

- An electric bulb is an electric device / a hollow spherical sealed glass bulb that consists in which a tungsten filament connected to two electrical contacts. The bulb is filled with unreactive gas like as Helium / Argon or Nitrogen when electric current passes through the filament, the filament has to be heated about 3000°C then electric bulb produces white light.



Electric Bulb

Note:

1. If Oxygen is filled in the bulb, then tungsten reacts with Oxygen to form tungsten Oxide which is brittle and break.
2. In the case of moisture tungsten also gets break.
3. The M.P of tungsten = 3400°C .

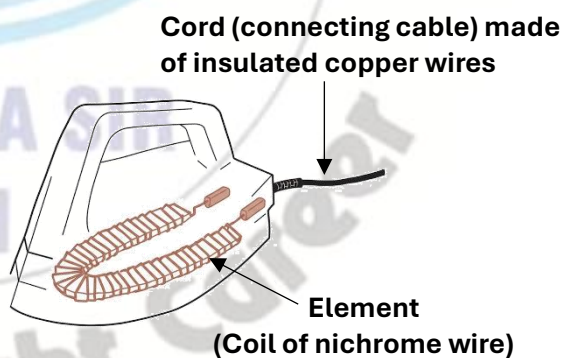
- **Rating of electrical appliances**

Q. An electric bulb is rated 100 w, 220 V. What is the meaning of this?

Ans If an electric bulb is rated 100 w, 220 V this means 220 V should be applied across this bulb the power consumed by this bulb is 100 w. i.e, 100 Joule per second consumed by this bulb.

- **Electric Iron**

- An electric Iron is that electric device which contain a high resistance coil of nichrome alloy when electricity passing through them a large amount of heat is produced in heating coil. It becomes red – hot, by conduction the connected plate gets hot which is used to ironing the clothes.



- **Electric heater**

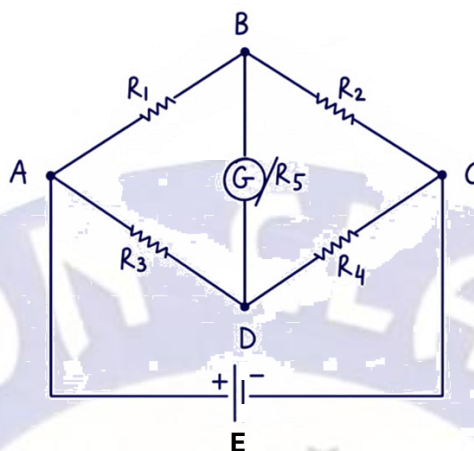
- An electric heater is a device that contain a high resistance coil of nichrome alloy. When electricity passes through it, a larger amount of heat is produced in the nichrome coil and becomes red – hot which is used to cook food, raise the temperature of room, car etc.



• **Wheat – Stone Bridge**

- The below figure shows the wheat's stone bridge.

Type 1:



- The bridge is most sensitive if all the four resistances R_1 , R_2 , R_3 and R_4 are of the same magnitude then we said that the bridge is balanced.

OR,

The bridge is said to be balanced if

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\Rightarrow R_1 \cdot R_4 = R_2 \cdot R_3$$

- In the Condition of balanced.

$$V_B = V_D$$

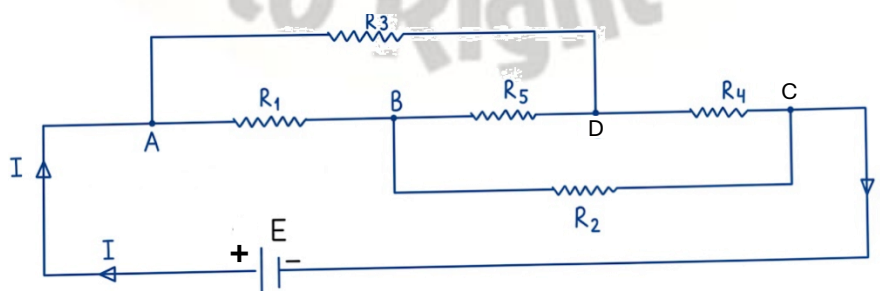
No current flows through galvanometer. The resistance of the galvanometer arm BD becomes ineffective and goes out of consideration.

- The cell – arm AEC and galvanometer arm BD are interchangeable. They are known as conjugated arms. When cell is connected between B & D, $V_A = V_C$ condition of balance point remains the same.

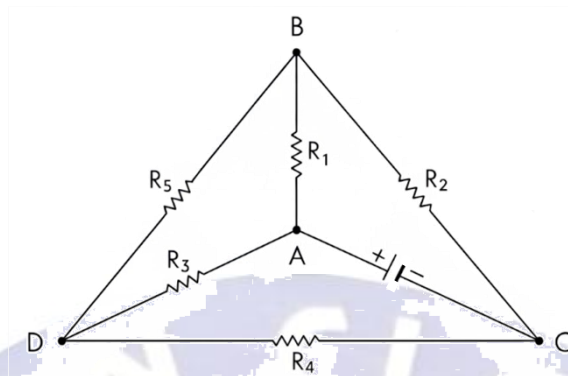
$$\Rightarrow R_1 \cdot R_4 = R_2 \cdot R_3$$

- The bridge is drawn in two more ways are depicted in below figure.

Type 2:

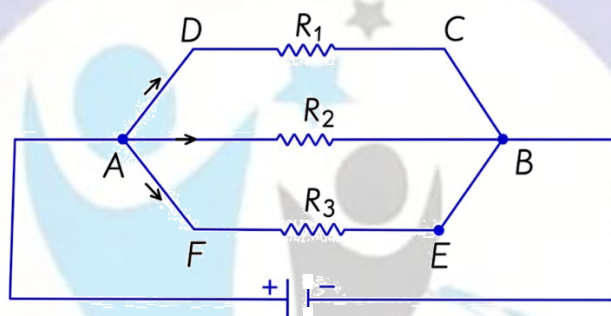


Type 3:



• **Junction**

- A junction in a circuit is a point where three or more conductors meet.
- Junctions are also called nodes / branch point.



Here, points A & B are called junctions.

• **Loop**

- A loop is a closed conducting path.

For example:-

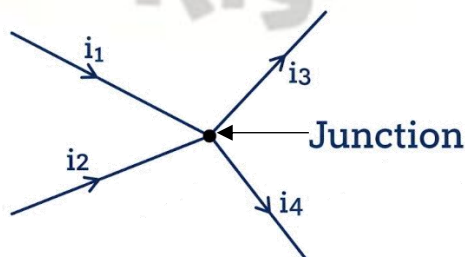
- ✓ ABCDA and ABEFA are called loop.

• **Kirchhoff's law**

1. **Kirchhoff's First law / Kirchhoff's Junction Rule**

- The algebraic sum of the electric current at any junction is zero.

i.e, $\sum_{\text{Junction}} i = 0$



i.e, $i_1 + i_2 = i_3 + i_4$

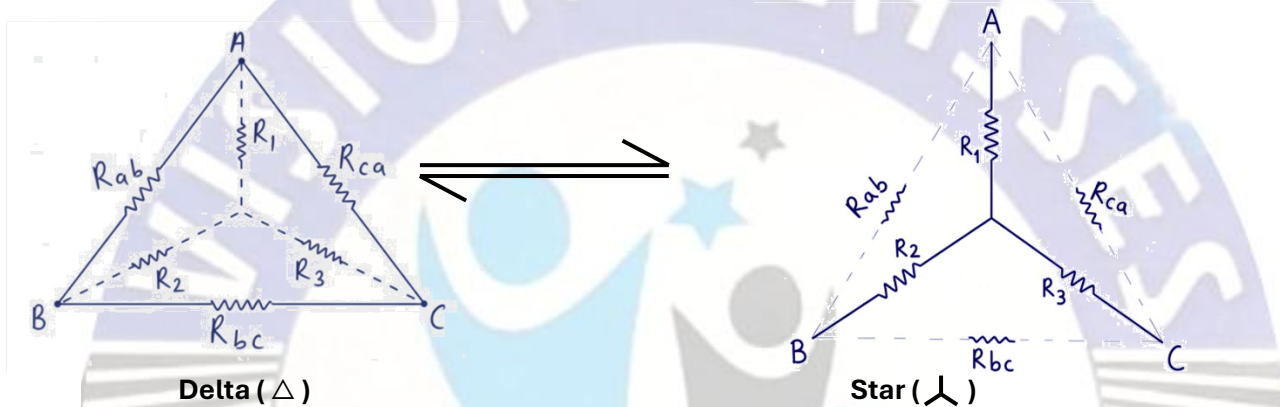
Note:

1. The junction rule is based on law of conservation of electric charge. So, no charge accumulates at a junction.
2. The entering current to be positive and leaving to be negative.
3. The algebraic sum of current at a junction must be zero.

2. Kirchhoff's Second law / Kirchhoff's loops rule

- The sum of electric potential difference in a closed loop is zero.
 i.e, $\sum_{\text{Closed loop}} \Delta V = 0$

• **Star and Delta Connection**



Case 1: Delta (Δ) to Star (Y)

$$R_1 = \frac{R_{ab} \times R_{ac}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_2 = \frac{R_{ab} \times R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

$$R_3 = \frac{R_{ac} \times R_{bc}}{R_{ab} + R_{bc} + R_{ca}}$$

Case 2: Star (Y) to Delta (Δ)

$$R_{ab} = R_1 + R_2 + \frac{R_1 \times R_2}{R_3}$$

$$R_{bc} = R_2 + R_3 + \frac{R_2 \times R_3}{R_1}$$

$$R_{ac} = R_1 + R_3 + \frac{R_1 \times R_3}{R_2}$$

Q. A metallic wire of resistance R is to be stretched and its length becomes double. Find the resistance of new wire.

Ans Let,

Length of metallic wire = l

Resistivity of metallic wire = ρ

Cross section area (thickness) = A

Since, Given Resistance of metallic wire = R

$$\therefore \rho = \frac{RA}{l} \quad \dots\dots\dots (i)$$

Now, Length of new wire = $2l$

$$\therefore \rho = \frac{R' \times \frac{A}{2}}{2l} \Rightarrow \rho = \frac{R' \times A}{4l} \quad \dots\dots\dots (ii)$$

From equation (i) and (ii)

$$\frac{R' \times A}{4l} = \frac{R \times A}{l}$$

$$\Rightarrow \frac{R'}{4} = R \Rightarrow R' = 4R$$

Q. A wire of resistance R_1 is cut into five equal pieces. These five pieces of wire are then connected in parallel. If the resultant resistance of this combination be R_2 , then the ratio $\frac{R_1}{R_2}$ is:

Ans A wire of resistance R_1 divides into five equal pieces

$$\therefore \text{Resistance of each piece} = \frac{R_1}{5}$$

$\therefore$ Case of 5 are connected in parallel

$$\therefore R_{eq} = \frac{r}{h} = \frac{\frac{R_1}{5}}{5} = \frac{R_1}{25}$$

$$\therefore R_2 = \frac{R_1}{25}$$

$$\therefore \frac{R_1}{R_2} = \frac{R_1}{\frac{R_1}{25}} = \frac{25}{1} = 25 : 1$$